1. create free API key at https://aistudio.google.com

2. Ollama to be installed → https://ollama.com/download

After ollama is installed (run this in cmd prompt)Pull models: `ollama pull gemma3:4b` and `ollama pull nomic-embed-text` (low-RAM machines: `gemma3:1b` instead)

3. Create Pinecone free account → https://www.pinecone.io

Python 3.13+ Installation Guide – Windows & macOS

Part 1: Windows Installation

Step 1: Check whether Python is already installed

1. Open Command Prompt.
    - Press Windows + R.
    - Type cmd.
    - Press Enter.
2. Execute the following commands one by one:
    - python --version
    - python3 --version
    - py --version
    - where python
    - py --list

If Python 3.13 or later is already installed, proceed to Step 5 to verify PATH and pip.

Step 2: Download Python for Windows

Official Python website

Download Python for Windows

1. Open the link above.
2. Find the latest stable Python 3.13 or newer release.
3. Download the Windows installer (64-bit) for most modern Windows computers.
4. Run the downloaded .exe installer.

Step 3: Install Python – Important configuration

Important: Do not skip this step

When the installer opens, you will see the installation screen.

Add Python to PATH on Windows 11

allthings.how

Before clicking Install Now:

1. Locate the checkbox at the bottom of the installer.
2. Select Add python.exe to PATH.
3. Make sure the checkbox is ticked.
4. Click Install Now.

This checkbox is important because it automatically adds Python to the Windows PATH environment variable.

Why is this necessary?

Without PATH configuration, you may need to provide the full Python installation path every time you execute a Python program.

With PATH configured, you can simply execute:

python --version

and run Python programs from any directory.

Step 4: Complete the installation

1. Wait until the installation is complete.
2. You should see a message indicating that the installation was successful.
3. Click Close.
4. Close Command Prompt if it was already open.
5. Open a new Command Prompt window.

Getting Started with Python

Step 5: Verify Python installation

Execute:

python --version

Expected output (example):

Python 3.13.13

Check pip:

python -m pip --version

If pip is not available, execute:

python -m ensurepip --upgrade

Check the Python installation location:

where python

You should see the location of the installed Python executable.

Step 6: If Python is installed but PATH is not configured

If Python is installed but python --version does not work, configure PATH manually.

1. Press the Windows key.
2. Search for Edit the system environment variables.
3. Open the option.
4. Click Environment Variables.
5. Under User variables, locate Path.
6. Select it and click Edit.
7. Click New.
8. Add the Python installation directory.

Typical paths (your version and installation location may differ):

C:\Users\YourUsername\AppData\Local\Programs\Python\Python313\

Add the Scripts directory as well:

C:\Users\YourUsername\AppData\Local\Programs\Python\Python313\Scripts\

9. Click OK on all open windows.
10. Close and reopen Command Prompt.
11. Execute:

python --version

Note: Do not copy these paths blindly. Use the actual installation location shown on your computer.

Part 2: macOS Installation

Unlike Windows, macOS does not normally provide an Add Python to PATH checkbox in the standard Python installer. We need to verify the installation and configure PATH if necessary.

Step 1: Check whether Python is already installed

1. Open Terminal.
    - Press Command + Space.
    - Type Terminal.
    - Press Enter.
2. Execute the following commands:

python3 --version

python3.13 --version

which python3

which -a python3

Important: macOS may already have a system-managed Python or another Python installation. Do not remove or modify the system-managed installation.

Step 2: Download Python for macOS

Official Python website

Download Python for macOSChoose a stable Python 3.13 or newer release.

1. Open the website.
2. Select a stable Python 3.13+ release.
3. Download the macOS installer (.pkg).
4. Open the downloaded .pkg file.

Step 3: Install Python

Follow these steps:

1. Double-click the downloaded .pkg file.
2. Click Continue.
3. Read and accept the software licence.
4. Select the installation destination.
5. Click Install.
6. Enter your Mac administrator password if requested.
7. Wait for the installation to complete.
8. Click Close.

The official installer normally creates command-line links so that Python can be accessed from Terminal.

There is no Windows-style PATH checkbox here.

Step 4: Verify Python installation

Open a new Terminal window and execute:

python3 --version

For a specific Python 3.13 installation:

python3.13 --version

Check pip:

python3 -m pip --version

Check the installation location:

which python3

If the output shows Python 3.13 or newer, the installation is successful.

Step 5: Configure PATH manually (only if required)

If Python is installed but Terminal cannot locate it, configure PATH.

Most modern macOS installations use the Zsh shell.

1. Open Terminal.
2. Execute:

nano ~/.zshrc

3. Add the following line at the bottom:

export PATH="/Library/Frameworks/Python.framework/Versions/3.13/bin:$PATH"

Replace 3.13 with your actual installed version if necessary.

4. Save the file:
    - Press Control + O.
    - Press Enter.
    - Press Control + X.
5. Reload the shell configuration:

source ~/.zshrc

6. Verify:

python3 --version

Important: Only add this path if you have installed Python using the official macOS installer and verified that this directory exists. If you installed through Homebrew, the PATH configuration will be different.

Part 3: Final Verification – Both Windows and macOS

Once installation is complete, perform the following checks.

Step 1: Verify Python version

Windows:

python --version

macOS:

python3 --version

Step 2: Verify pip

Windows:

python -m pip --version

macOS:

python3 -m pip --version

Step 3: Create a test Python program

Create a file named test_python.py.

Add the following code:

import sysprint("Python Installation Successful!")print("Python Version:", sys.version)print("Python Executable:", sys.executable)

Execute it:

Windows:

python test_python.py

macOS:

python3 test_python.py

You should see the Python version and the installation location.